

Peripherals, Interfacing and IoT (Introduction)

Introduction

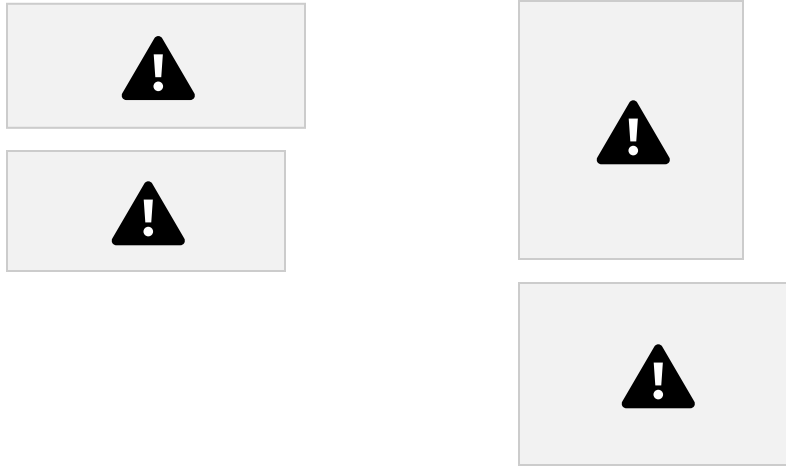


Computer Systems

- Internal (processor + memory (RAM)) – Peripheral (Disk, Display, Audio, Eth,..)

Introduction





- Peripherals : HD monitor, 5.1 speaker •
- Interfaces : Intermediate Hardware – Nvidia GPU card, Creative Sound Blaster card •
- Interfaces : Intermediate Software/Program

- Nvidia GPU driver , Sound Blaster Driver software

Introduction

- Interfaces
 - Intermediate Hardware
 - Timer, Counter, DMA, USB
 - Peripheral Controller
 - Intermediate Software/Program
- Device driver (Linux)/ Assembly Code
- Peripheral Component

Interconnect(PCI)

– Audio card, VGA card, Ethernet card • Low level signal + high level C code

Linux Kernel Split View



Linux Device Driver by Jonhantan Corbet

Motivation

- Knowledge: both hardware & software
- Exact interface: Architecture & OS
- Used in many places (Computer +ES)
 - All embedded system (mobile, laptop, ..)

- Highly paid job in industries
 - Intel (BIOS, driver), Sony, Motorola, ..
- Low level signal + Device drivers

Motivation Contd..

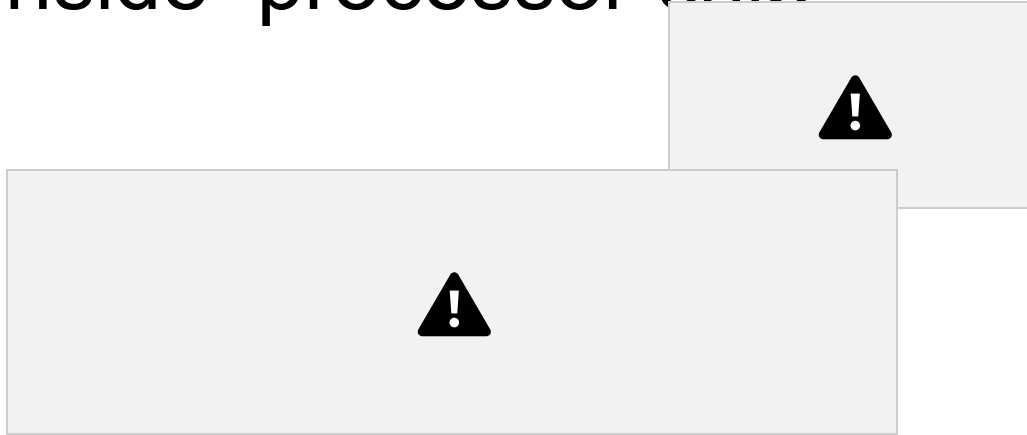
- Knowledge of simple peripherals (Display, Audio, Disk drives, Ethernet) In connection with 8085/8086

- Peripheral are powerful than main computing, knowledge of advance peripherals
 - Linux/Window device drivers
 - Dolby Digital Stereo, HD Cinema – Most influential technology of 2010 • USB 3.0, Bluetooth 4
- Graphics cards (Nvidia with 480 core)

Motivation Contd..

- Use of old technology in newer devices
 - Intel Atom processor (PII technology with modification)
 - Use of winXP in mobile; may be obsolete for PC
- Combining peripheral controller in main computing for low power
 - Intel Centrino have wireless controller functionality inside processor chip
 - Intel atom 45x have DDR2 memory

controller + Graphics controller in
inside processor chip



Dolby Digital

- Stereo: Moving origin of sound in a line – Stereo Ear Phone:
 - Stereo Image: to calculate Depth :

why two eye in human face?

- Dolby Lab: 5.1 (5 normal + 1 subwoofer) – 5 normal : 20-20Khz, 1 low freq (20-120K) – RFront, Centre, LFrnt, RSorround, Lsorrnd – Max bit rate: 560 bit/s
- Dolby HD : 7.1, Max bit rate: 18MBs

HD Cinema

- Video : 30 frame/ Sec

- 1 Hr Video size with out compression
 - (Resolution).(3 color).(2byte).(30F/S).60M.60S –
 - VGA:640x480: 199GB , Comp: 450 Mb –
 - 720p:1280x720: 597GB, Comp:1.2Gb
 - 1080p/i:1920x1080: 1.35TB, Comp:2.4G •
- MP2, MP4, MKV matryoshka (nested doll) •
- Cinema:Old-2K(2048×1080),New- 4K(4096×2160) •
- TI- Digital Light Processing, Sony: SXRD (Silicon X-tal Reflective Display), LCOS (liquid crystal on silicon)

MI Tech 2010 :Nvidia

GPUs



• Nvidia

GTX295, 480 Cuda core

- VGA upto Res: 2048x1536, Dual monitor
- HD Cinema, Play MKV File

MI Tech 2010 : USB 3.0

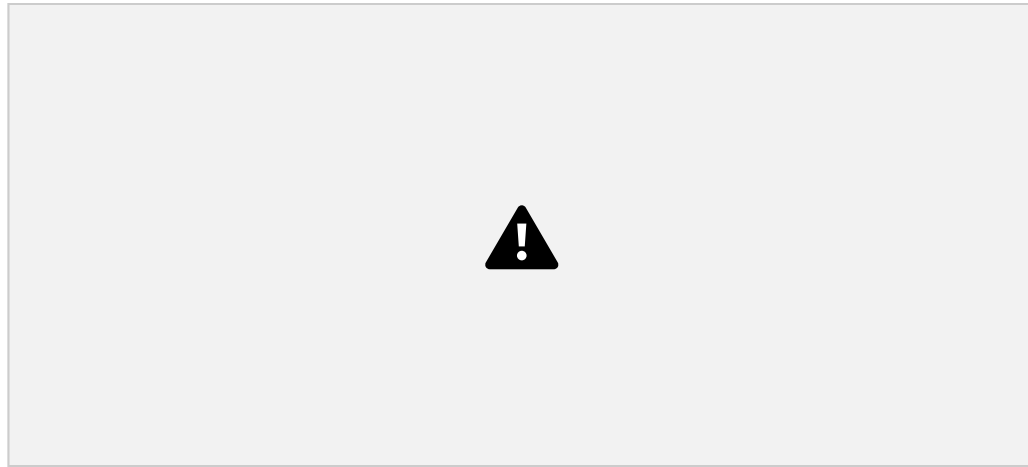
- SuperSpeed" bus
 - Enhanced Host Controller Interface (EHCI)
 - Register-level interface: Host Controller for the Universal Serial Bus (USB2.0)
 - SATA HDD: Serial Adv. Tech. Attachment. •
- USB3.0: Transfer mode at 5.0 Gbit/s

=400MB/s • It uses

- 8B/10B encoding, LinearFBshftReg (LFSR) scrambling for data, Spread Spectrum.
- Receivers: low freq periodic signaling, dynamic equalization & training sequences

MI Tech 2010: Bluetooth

4.0



Classic Bluetooth

- Radio: Freq Hop Spread Spectrum, 2.4Ghz –1Meters, 3MB/s
- Bluetooth high speed (based on WiFi)
- Bluetooth low energy (Added

4.0 Spec)

Peripherals Controller **Migration**



- Cards on Mother board

- Onboard: graphics, modem, audio, Wireless
- Inside processor: graphics, memory cont.

Intel Centrino Processor





• 5X

better wireless performance with
Proactive

Security (Added Instruction to support this) • Longer Battery Life (Low power instr & FUs)

Intel Atom Processor



- Old Pentium architecture with modification (Low power addition) : 10Watt

- Default: MMX, SSE (Streaming SIMD)
- Inside processor chip
 - Graphics processor, Memory controller – Wireless controller (Centrino Atom)

Book

- Text
 - R S Gaonkar, “Microprocessor Architecture, Programming and Application with the 8085”, 5th edition, Penram India

- J. Corbet, A Rubini, “Linux Device Driver”
3rd Edition, O’relly publisher

- **References**

- D V Hall, Microprocessors and Interfacing, TMH, 1995 – M B Cook and H White Neil, Computer Peripherals, 3/e. London: Edward Arnold, 1995.
- L F Doyle, Computer Peripherals, Prentice Hall, 19

Thanks